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Material Safety Datasheet

Issue date: 25/10/2016

Section 1 - Product and Company Identification

Product Name

Triethanolamine, Commercial Grade

COMPANY IDENTIFICATION

Dow Chemical Thailand Ltd
A Subsidiary of The Dow Chemical Company
15th floor, White Group II
75 Soi Rubia, Sukhumvit 42
Prakanong, Bangkok 10110
Thailand

Customer Information Number: 02-3657000
SDSQuestion@dow.com

EMERGENCY TELEPHONE NUMBER

24-Hour Emergency Contact: 038-925400
Local Emergency Contact: (66) 38 925400

Section 2 - Hazards Identification

Emergency Overview
Color: Colorless to yellow
Physical State: Liquid.
Odor: Ammoniacal
Hazards of product:

WARNING! Causes eye irritation. May cause skin irritation. Isolate area.

Potential Health Effects
Eye Contact: May cause moderate eye irritation. May cause moderate corneal injury.
Skin Contact: Prolonged contact may cause skin irritation with local redness. Repeated contact may cause skin burns. Symptoms may include pain, severe local redness, swelling, and tissue damage.
Skin Absorption: Prolonged skin contact is unlikely to result in absorption of harmful amounts.
Skin Sensitization: For the major component(s): Skin contact may cause an allergic skin reaction in a small proportion of individuals.
Inhalation: At room temperature, exposure to vapor is minimal due to low volatility. Vapor from heated material may cause respiratory irritation and other effects.
Ingestion: Low toxicity if swallowed. Small amounts swallowed incidentally as a result of normal handling operations are not likely to cause injury; however, swallowing larger amounts may cause injury.
Effects of Repeated Exposure: Based on information for component(s): In animals, effects have been reported on the following organs: Kidney. Liver. Blood. Results from repeated exposure tests on diethanolamine in laboratory animals include anemia (rats) and effects on kidney (rats and mice) and liver (mice). Heart and nervous system effects were also observed in animals given exaggerated dose of diethanolamine. Changes in other organs, causes of which are nonspecific, were judged secondary to the poor health of the animals due to the extremely high doses of diethanolamine given.
Cancer Information: For the major component(s): Findings from a chronic diethanolamine skin painting study by NTP include liver and kidney tumors in mice; no tumors were observed in rats. Mechanistic studies indicate that tumor formation is of questionable relevance to humans. A number of factors may have influenced the results and are being considered in their interpretation. Findings from a chronic skin painting study by NTP include liver tumors in mice. Mechanistic studies indicate that tumor formation of questionable relevance to humans.
Birth Defects/Developmental Effects: Contains component(s) which did not cause birth defects in animals; other fetal effects occurred only at doses toxic to the mother.

Section 3 - Composition Information

Component	CAS #	Amount
Triethanolamine	102-71-6	85.0 - 89.0 %
N,N-Diethanolamine	111-42-2	11.0 - 15.0 %

Section 4 - First-aid measures

Eye Contact: Flush eyes thoroughly with water for several minutes. Remove contact lenses after the initial 1-2 minutes and continue flushing for several additional minutes. If effects occur, consult a physician, preferably an ophthalmologist.
Skin Contact: Wash skin with plenty of water.
Inhalation: Move person to fresh air, it effects occur, consult a physician
Ingestion: If swallowed, seek medical attention. Do not induce vomiting unless directed to do so by medical personnel.
Notes to Physician: If burn is present, treat as any thermal burn, after decontamination. No specific antidote. Treatment of exposure should be directed at the control of symptoms and the clinical condition of the patient.
Emergency Personnel Protection: First Aid responders should pay attention to self-protection and use the recommended protective clothing (chemical resistant gloves, splash protection). If potential for exposure exists refer to Section 8 for specific personal protective equipment.

Section 5 - Fire Fighting Measures

Extinguishing Media: Water fog or fine spray. Dry chemical fire extinguishers. Carbon dioxide fire extinguishers. Foam. Do not use direct water stream. May spread fire. Alcohol resistant foams (ATC type) are preferred. General purpose synthetic foams (including AFFF) or protein foams may function, but will less effective.
Fire Fighting Procedures: Keep people away. Isolate fire and deny unnecessary entry. Use water spray to cool fire exposed containers and fire affected zone until fire is out and danger of reignition has passed. Fight fire from protected location or safe distance. Consider the use of unmanned hose holders or monitor nozzles. Immediately withdraw all personnel from the area in case of rising sound from venting safety device or discoloration of the container. Burning liquids may be extinguished by dilution with water. Do not use direct water stream. May spread fire. Move container from fire area if this is possible without hazard. Burning liquids may be moved by flushing with water to protect personnel and minimize property damage.
Special Protective Equipment for Firefighters: Wear positive-pressure self-contained breathing apparatus (SCBA) and protective fire fighting clothing (includes fire fighting helmet, coat, trousers, boots, and gloves). Avoid contact with this material during fire fighting operations. If contact is likely, full chemical resistant fire fighting clothing with self-contained breathing apparatus. If this is not available, wear full chemical resistant clothing with self-contained breathing apparatus and fight fire from a remote location. For protective equipment in post-fire or non-fire clean-up situations, refer to the relevant sections.
Unusual Fire and Explosion Hazards: Container may rupture from gas generation in a fire situation. Violent steam generation or eruption may occur upon application of direct water stream to hot liquids.
Hazardous Combustion Products: During a fire, smoke may contain the original material in addition to combustion products of varying composition which may be toxic and/or irritating. Combustion products may include and are not limited to: Nitrogen oxides. Carbon monoxide. Carbon dioxide.

Section 6 - Accidental Release Measures

Steps to be Taken if Material is Released or Spilled: Contain spilled material if possible. Small spills: Absorb with materials such as: Non-combustible material. Clay. Vermiculite. Zorb-all@. Large spills: Pump into suitable and properly labeled containers. See Section 13, Disposal Considerations, for additional information.
Personal Precautions: Isolate area. Keep unnecessary and unprotected personnel from entering the area. Refer to Section 7, Handling, for additional precautionary measures. Use appropriate safety equipment. For additional information, refer to Section 8,Exposure Controls and Personal Protection.
Environmental Precautions: Prevent from entering into soil, ditches, sewers, waterways and/or groundwater. See Section 12, Ecological Information.

Section 7 - Handling and Storage

Handling
General Handling: Avoid contact with eyes, skin, and clothing. Wash thoroughly after handling. Keep container closed. Use with adequate ventilation. Do not use sodium nitrite or other nitrosating agents in formulations containing this product. Suspected cancer-causing nitrosamines could be formed. Spills of these organic materials on hot fibrous insulations may lead to lowering of the autoignition temperatures possibly resulting in spontaneous combustion. See Section 8, EXPOSURE CONTROLS AND PERSONAL PROTECTION.
Storage
Store in a dry place. Avoid freezing. Thaw and mix well before using. Store in the following material(s): Stainless steel. Do not store in: Galvanized steel. Copper. Copper alloys. Zinc.

Storage Period:

Bulk

6 Months

Steel drums.

24 Months

Section 8 - Exposure Controls / Personal Protection

Exposure Limits

Component	List	Type	Value
Triethanolamine	ACGIH	TWA	5 mg/m3
N,N-Diethanolamine	ACGIH	TWA	1 mg/m3 SKIN
		Inhalable fraction and vapor.	

A "skin" notation following the inhalation exposure guideline refers to the potential for dermal absorption of the material including mucous membranes and the eyes either by contact with vapors or by direct skin contact. It is intended to alert the reader that inhalation may not be the only route of exposure and that measures to minimize dermal exposures should be considered.

Personal Protection

Eye\Face Protection: Use chemical goggles.

Skin Protection: When prolonged or frequently repeated contact could occur, use protective clothing chemically resistant to this material. Selection of specific items such as faceshield, boots, apron, or full-body suit will depend on the task.

Hand protection: Use gloves chemically resistant to this material when prolonged or frequently repeated contact could occur. Examples of preferred glove barrier materials include: Chlorinated polyethylene. Polyethylene. Ethyl vinyl alcohol laminate ("EVAL"). Examples of acceptable glove barrier materials include: Butyl rubber. Natural rubber ("latex"). Neoprene. Nitrile/butadiene rubber ("nitrile" or "NBR"). Polyvinyl chloride ("PVC" or "vinyl"). Viton. Avoid gloves made of: Polyvinyl alcohol ("PVA"). NOTICE: The selection of a specific glove for a particular application and duration of use in a workplace should also take into account all relevant workplace factors such as, but not limited to: Other chemicals which may be handled, physical requirements (cut/puncture protection, dexterity, thermal protection). potential body reactions to glove materials, as well as the instructions/specifications provided by the glove supplier.

Respiratory Protection : Respiratory protection should be worn when there is a potential to exceed the exposure limit requirements or guidelines. there are no applicable exposure limit requirements or guidelines, wear respiratory protection when adverse effects, such as respiratory irritation or discomfort have been experienced, or where indicated by your risk assessment process. For most conditions, no respiratory protection should be needed; however, if handling at elevated temperatures without sufficient ventilation, use an approved air-purifying respirator. The following should be effective types of air-purifying irators: Organic vapor cartridge with a particulate pre-filter.

Ingestion: Use good personal hygiene. Do not consume or store food in the work area. Wash hands before smoking or eating.

Engineering Controls

Ventilation: Use local exhaust ventilation, or other engineering controls to maintain airborne levels below exposure limit requirements or guidelines. If there are no applicable exposure limit requirements or guidelines, general ventilation should be sufficient for most operations. Local exhaust ventilation may be necessary for some operations.

Section 9 - Physical and Chemical Properties

Physical State	Liquid.
Color	Colorless to yellow
Odor	Ammoniacal
Odor Threshold	No test data available
Flash Point - Closed Cup	194.4°C ASTM D93
Flash Point - Open Cup	190.5 °C ASTM D92
Flammability (solid, gas)	No
Flammable Limits In Air	**Lower:** Not determined **Upper:** Not determined
Autoignition Temperature	No test data available
Vapor Pressure	< 0.01 mmHg @ 20 °C Literature
Boiling Point (760 mmHg)	310.3 °C Estimated. Extrapolated.
Vapor Density (air = 1)	4.9 Literature
Specific Gravity (H20 = 1)	1.126 Unspecified
Freezing Point	15.8 °C Literature
Melting Point	Not applicable
Solubility in water (by weight)	100% @ 20 °C Literature
pH	Not available
Decomposition Temperature	No test data available
Partition coefficient, n-octanol/water (log Pow)	No test data available
Evaporation Rate (ButylAcetate = 1)	<0.01 Not reported
Kinematic Viscosity	No test data available

Section 10 - Stability and Reactivity

Stability/Instability
Stable under recommended storage conditions. See Storage, Section 7. Hygroscopic.
Conditions to Avoid: Exposure to elevated temperatures can cause product to decompose. Generation of gas during decomposition can cause pressure in closed systems. Avoid moisture.

Incompatible Materials: Avoid contact with: Nitrites. Strong acids. Strong oxidizers. Product may potentially react with various halogenated organic solvents, resulting in temperature and/or pressure increases Corrosive when wet. Heating above 60°C in the presence of aluminum can result in corrosion and generation of flammable hydrogen gas. Avoid unintended contact with: Halogenated hydrocarbons.
Hazardous Polymerization
will not occur.
Thermal Decomposition
Decomposition products depend upon temperature, air supply and the presence of other materials.

Section 11 Toxicological Information

Acute Toxicity
Ingestion
For similar material(s): LD50, Rat > 4,000 mg/kg
Skin Absorption
Based on information for component(s): LD50, Rabbit > 2,000 mg/kg
Inhalation
As product. The LC50 has not been determined.
Sensitization
Skin
For the major component(s): Skin contact may cause an allergic skin reaction in a small proportion of individuals. For the major component(s): Did not cause allergic skin reactions when tested in guinea pigs.
Repeated Dose Toxicity
Based on information for component(s): In animals, effects have been reported on the following organs: Kidney. Liver. Blood. Results from repeated exposure tests on diethanolamine in laboratory animals include anemia (rats) and effects on kidney (rats and mice) and liver (mice). Heart and nervous system effects were also observed in animals given exaggerated doses of diethanolamine. Changes in other organs, causes of which are nonspecific, were judged secondary to the poor health
of the animals due to the extremely high doses of diethanolamine given.
Chronic Toxicity and Carcinogenicity
For the major component(s): Findings from a chronic diethanolamine skin painting study by NTP include liver and kidney tumors in mice; no tumors were observed in rats. Mechanistic studies indicate that tumor formation is of questionable relevance to humans. A number of factors may have influenced the results and are being considered in their interpretation. Findings from a chronic skin painting study by NTP include liver tumors in mice. Mechanistic studies indicate that tumor formation is of questionable relevance humans.
Carcinogenicity Classifications:

Component	List	Classification
N,N-Diethanolamine	ACGIH	Confirmed animal carcinogen with unknown relevance to humans ; Group A3

Developmental Toxicity
Contains component(s) which did not cause birth defects in animals; other fetal effects occurred only at doses toxic to the mother. For the major component(s): Triethanolamine. Screening studies in animals suggest that this material does not affect fetal development.
Reproductive Toxicity
No relevant information found.
Genetic Toxicology
Contains a component(s) which were negative in in vitro genetic toxicity studies. For the minor component(s): Animal genetic toxicity studies were negative.

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Section 12 - Ecological Information

ENVIRONMENTAL FATE

Data for Component: **Triethanolamine**

Movement & Partitioning

Bioconcentration potential is low (BCF less than 100 or log Pow less than 3). Potential for mobility in soil is very high (Koc between 0 and 50).

Henry's Law Constant (H): 1.91E-04 atm*m3/mole Measured

Partition coefficient, n-octanol/water (log Pow): -1.00 Measured

Partition coefficient, soil organic carbon/water (Koc): 3 Estimated.

Bioconcentration Factor (BCF): < 3.9; common carp (Cyprinus carpio); Measured

Distribution in Environment: Mackay Level 1 Fugacity Model:

Air	Water.	Biota	Soil	Sediment
	> 99 %			

Persistence and Degradability

Material is readily biodegradable. Passes OECD test(s) for ready biodegradability. Material is ultimately biodegradable (reaches > 70% mineralization in OECD test(s) for inherent biodegradability).

Indirect Photodegradation with OH Radicals

Rate Constant	Atmospheric Half-life	Method
1.1053E-10 cm3/s	0.097 d	Estimated.

OECD Biodegradation Tests:

Biodegradation	Exposure Time	Method
97 %	28 d	OECD 301 A Test
89 %	14 d	OECD 302B Test
92 %	3 h	OECD 303A Test

Theoretical Oxygen Demand: 2.04 mg/mg

Data for Component: **N.N-Diethanolamine**

Movement & Partitioning

Bioconcentration potential is low (BCF less than 100 or log Pow less than 3). Potential for mobility in soil is very high (Koc between 0 and 50). Given its very low Henry's constant, volatilization from natural bodies of water or moist soil is not expected to be an important fate process.

Henry's Law Constant (H): 5.35E-14 atm*m3/mole; 25 °C Measured

Partition coefficient, n-octanol/water (log Pow): -2.18 Shake flask (OECD 107 Test)

Partition coefficient, soil organic carbon/water (Koc): 1 Estimated.

Persistence and Degradability

Material is readily biodegradable. Passes OECD test(s) for ready biodegradability. Material is ultimately biodegradable (reaches > 70% mineralization in OECD test(s) for inherent biodegradability)

Indirect Photodegradation with OH Radicals

Section 12 - Ecological Information

Rate Constant	Atmospheric Half-life	Method
8.90E-11 cm3/s	0.167 d	Estimated.

OECD Biodegradation Tests:

Biodegradation	Exposure Time	Method
100 %	19 d	OECD 301 E Test

Theoretical Oxygen Demand: 2.13 mg/mg

ECOTOXICITY

Data for Component: **Triethanolamine**

Material is practically non-toxic to aquatic organisms on an acute basis (LC50/EC50/EL50/LL50 >100 mg/L in the most sensitive species tested). May increase pH of aquatic systems to > pH 10 which may be toxic to aquatic organisms.

Fish Acute & Prolonged Toxicity

LC50, fathead minnow (Pimephales promelas), 96 h: 1,800 - 11,800 mg/l

Aquatic Invertebrate Acute Toxicity

LCD50, water flea Daphnia magna, 24 h: 739 - 2,038 mg/l

Aquatic Plant Toxicity

EC50, alga Scenedesmus sp., Growth rate inhibition, 72 h: 216 - 750 mg/

Toxicity to Micro-organisms

EC50, OECD 209 Test; activated sludge, respiration inhibition, 3 h: > 1,000 mg/l

Data for Component: **N.N-Diethanolamine**

Material is moderately toxic to aquatic organisms on an acute basis (LC50/EC50 between 1 and 10 mg/L in the most sensitive species tested). May increase pH of aquatic systems to > pH 10 which may be toxic to aquatic organisms.

Fish Acute & Prolonged Toxicity

LC50, bluegill (Lepomis macrochirus), static: 1,850 - 2,100 mg/l

Aquatic Invertebrate Acute Toxicity

LC50, water flea Daphnia magna, 48 h: 122 mg/l

Aquatic Plant Toxicity

EC50, green alga Pseudokirchneriella subcapitata (formerly known as Selenastrum capricornutum), biomass growth inhibition, 96 h: 3.3 - 3.6 mg/l

Toxicity to Micro-organisms

EC50, OECD 209 Test: activated sludge, respiration inhibition, 3 h: > 1,000 mg/l

Section 13 - Disposal Considerations

DO NOT DUMP INTO ANY SEWERS, ON THE GROUND, OR INTO ANY BODY OF WATER. All disposal practices must be in compliance with all Federal, State/Provincial and local laws and regulations. Regulations may vary in different locations. Waste characterizations and compliance with applicable laws are the responsibility solely of the waste generator. AS YOUR SUPPLIER, WE HAVE NO CONTROL OVER THE MANAGEMENT PRACTICES OR MANUFACTURING PROCESSES OF PARTIES HANDLING OR USING THIS MATERIAL. THE INFORMATION PRESENTED HERE PERTAINS ONLY TO THE PRODUCT AS SHIPPED IN ITS INTENDED CONDITION AS DESCRIBED IN MSDS SECTION: Composition Information. FOR UNUSED & UNCONTAMINATED PRODUCT, the preferred options include sending to a licensed, permitted: Incinerator or other thermal destruction device.

Section 14 - Transport Information

ROAD & RAIL Non-Bulk
NOT REGULATED
ROAD & RAIL Bulk
NOT REGULATED
IMDG
II NOT REGULATED
ICAO/IATA
NOT REGULATED
This information is not intended to convey all specific regulatory or operational requirements/information relating to this product. Additional transportation system information can be obtained through an authorized sales or customer service representative. It is the responsibility of the transporting organization to follow all applicable laws, regulations and rules relating to the transportation of the material.

Section 15 - Regulatory Information

Thailand: Hazardous Substance Act, B.E. 2535
A component(s) of this product is classified according to Thai Hazardous Substance Acts-type
3.Thailand: Notification of Ministry of Interior (Occupational Health and Safety in Chemical).
One or more components of this product are listed.

Section 16 - Other Information

Product Literature
Additional information on this product may be obtained by calling your sales or customer service contact. Ask for a product brochure. Additional information on this and other Dow products may be obtained by visiting our web page at www.dow.com

Recommended Uses and Restrictions
Chemical intermediate. Chemical additive. We recommend that you use this product in a manner consistent with the listed use. If your intended use is not consistent with the stated use, please contact your sales or technical service representative.

Revision
Identification Number: 2138/4073/ Issue Date 10/02/2009/ Version: 3.0
Most recent revision(s) are noted by the bold, double bars in left-hand margin throughout this document.

Legend

N/A	Not available
W/W	Weight/Weight
OELO	ccupational Exposure Limit
STEL	Short Term Exposure Limit
TWA	Time Weighted Average
ACGIH	American Conference of Governmental Industrial Hygienists, Inc.
DOW IHG	Dow Industrial Hygiene Guideline
WEEL	Workplace Environmental Exposure Level
HAZ_DES	Hazard Designation

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